IE 7615- Deep Learning for AI

DISCRIMINATIVE PROJECT MILESTONE 2 REPORT

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1. Dataset and Problem Overview

Milestone 1 classified a photo containing a single object. In this milestone we move to object detection: given an image that contains several of the class's objects, the model has to report the Object ID and the location (bounding box) of every object in it. We used the final merged class dataset and the final Object IDs from the shared sheet.

Following the course announcement, OBJ054 was excluded because its photos do not show the object, which leaves 72 Object IDs (OBJ001 to OBJ073 without OBJ054). Byte-identical duplicate photos were removed before splitting, the one OBJ003 photo that is not 224 × 224 is resized when it is placed in a grid, and background-only photos (*_bg_*) are not used. We kept the stratified 70/15/15 split from Milestone 1 (seed 42), and every grid image is built only from photos of one split, so no source photo appears in more than one of train, validation and test (0 overlapping photos across splits).

2. Bounding-Box Labelling and Label Checking

The shared dataset only gives each photo a class through its folder name, with no bounding box, so we created the boxes automatically with open-vocabulary detectors. Each detector was prompted with the object's name from the sheet, plus a clearer everyday phrase where the sheet name was unusual (for example "hair claw clip" for Clutcher, "ping pong paddle" for Table Tennis Racket). We ran the detectors at the photos' native 224 × 224 size, since upscaling the photos made them miss most objects in our tests. Three models were used:

- OWLv2 (google/owlv2-base-patch16-ensemble), the primary labeller and the strongest of the three on our photos.
- YOLO-World (yolov8s-worldv2) and YOLOE (yoloe-11s-seg), used as independent second opinions.

A box was accepted only under the following rules, and every other photo was left out:

- Verified: OWLv2's box agrees with the YOLO-World or YOLOE box (IoU ≥ 0.5). OWLv2's box is used.
- Two-model: OWLv2 found nothing, but YOLO-World and YOLOE agree with each other (IoU ≥ 0.5). Their average box is used.
- Single-model: only OWLv2 found the object, with a confidence of at least 0.35. These boxes were spot-checked by eye.
- Excluded: the models disagree, no model finds the object, or the box covers less than 2% of the photo (object far away or in heavy clutter).

Label status	Photos	Share
Kept: verified (OWLv2 + a second model)	3,947	57.5%
Kept: two-model (YOLO-World + YOLOE)	93	1.4%
Kept: single-model (OWLv2 only)	1,250	18.2%
Left out: models disagree	866	12.6%
Left out: no model found the object	375	5.5%
Left out: object too small (< 2% of photo)	328	4.8%
Kept in total	**5,290**	**77.1%**

Out of 6,859 photos, 5,290 received a box (77%) and 1,569 were left out. Every one of the 72 objects kept at least 13 labelled photos; the objects with the fewest were Sleep Mask (OBJ060) with 13, Mouth guard case (OBJ042) with 24, NoteBook (OBJ003) with 34, RGB Projector (OBJ064) with 37. These are mostly small, thin or flat objects that the detectors find hard to separate from the surface they are lying on.

We then checked a random sample of 64 kept boxes by eye. 63 were correct, meaning the box covered the labelled object and not something else in the photo. The sample had 32 verified and 32 single-model boxes. The one imperfect box (single-model, Nike sneakers OBJ010) covered only one shoe of the pair; none of the 64 boxes was on the wrong item. Single-model boxes are slightly looser on average but still enclose the object.

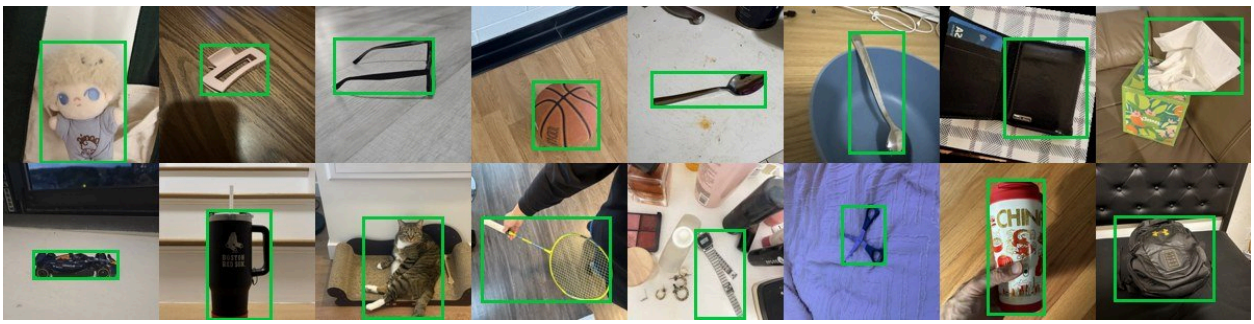


Figure 1. Random sample of kept labels (green = final bounding box).



Figure 2. Photos left out because the models disagree (red = OWLv2, blue = YOLO-World, orange = YOLOE).

3. Multi-Object Grid Images

We built the multi-object images by placing 224 × 224 photos side by side in a grid, using all five grid sizes from the course specification in equal proportion. Every cell holds a different object, chosen at random from the photos of that split that have a kept label, and its bounding box is the photo's labelled box shifted into the cell. Each cell is augmented with a random horizontal flip (the box is mirrored with it) and brightness, contrast and saturation jitter.

Grid	Image size	Objects / image	Train	Val	Test	Test mAP@0.5 (YOLOv8s)	Test recall (YOLOv8s)
2x2	448 × 448	4	200	30	30	0.828	0.755
3x3	672 × 672	9	200	30	30	0.920	0.859
4x4	896 × 896	16	200	30	30	0.878	0.834
5x4	1120 × 896	20	200	30	30	0.899	0.858
5x5	1120 × 1120	25	200	30	30	0.915	0.869
Total			**1000**	**150**	**150**		

In total the group generated 1,300 grid images containing 19,240 labelled objects (14,800 train, 2,220 validation, 2,220 test), drawn from 3,617, 716 and 733 distinct source photos respectively.

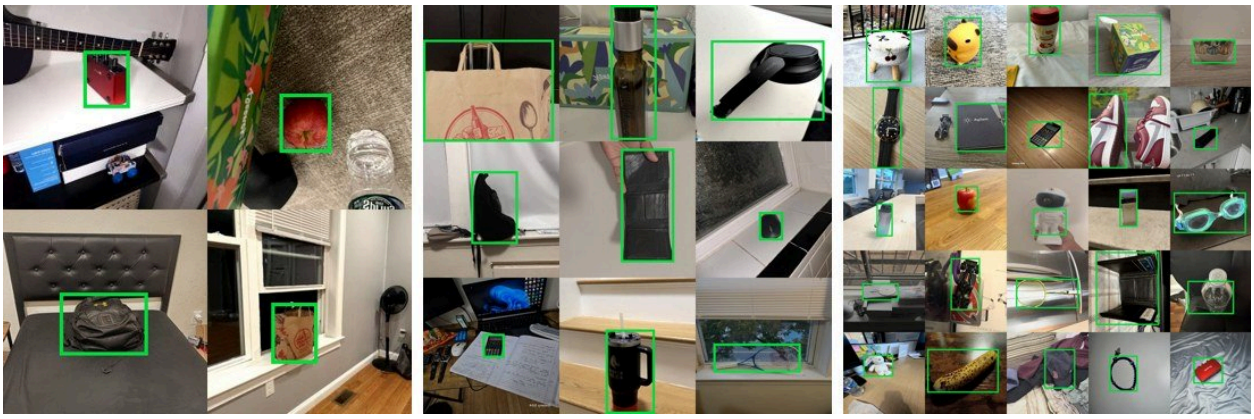


Figure 3. Generated 2×2, 3×3 and 5×5 training grids with their bounding boxes.

4. Model and Training Setup

We fine-tuned two sizes of YOLOv8 from COCO-pretrained weights (transfer learning), YOLOv8n (nano) and YOLOv8s (small), with the detection head re-initialised for our 72 classes.

- Image size: 1120, matching the largest grid, so objects in 5×5 grids are not shrunk
- Batch size: 8 for YOLOv8n; 4 for YOLOv8s, which did not fit in memory at batch 8 with 1120-pixel images
- Optimizer and learning-rate schedule: Ultralytics defaults (automatic optimizer selection with warm-up)
- Up to 50 epochs for YOLOv8n and 30 for YOLOv8s (the larger model is about twice as slow per epoch), early stopping with patience 15; the best epoch by validation fitness is kept
- Training-time augmentation: mosaic (50%, turned off for the last 10 epochs), HSV jitter, horizontal flip, scale and translation
- Checkpoints saved every 5 epochs; trained on an Apple M4 GPU (PyTorch MPS)

5. Experimental Results

Results on the held-out test set (150 grid images, 2,220 objects):

Model	mAP@0.5	mAP@0.5:0.95	Precision	Recall	Params	Epochs	Train (min)	Infer. (ms/img)
YOLOv8s	0.8934	0.8162	0.8901	0.8557	11.16M	30	149	4.6
YOLOv8n	0.8769	0.7793	0.8719	0.8319	3.09M	50	116	3.9

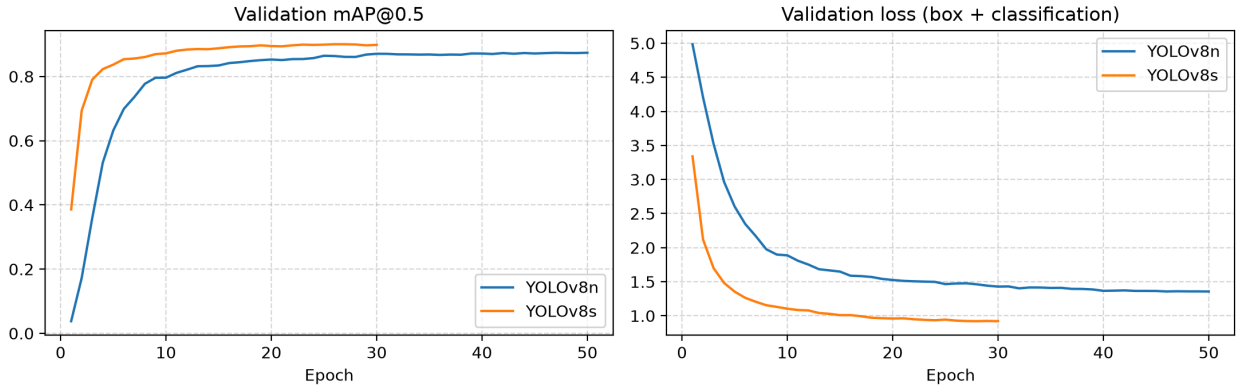


Figure 4. Validation mAP@0.5 and validation loss per epoch.

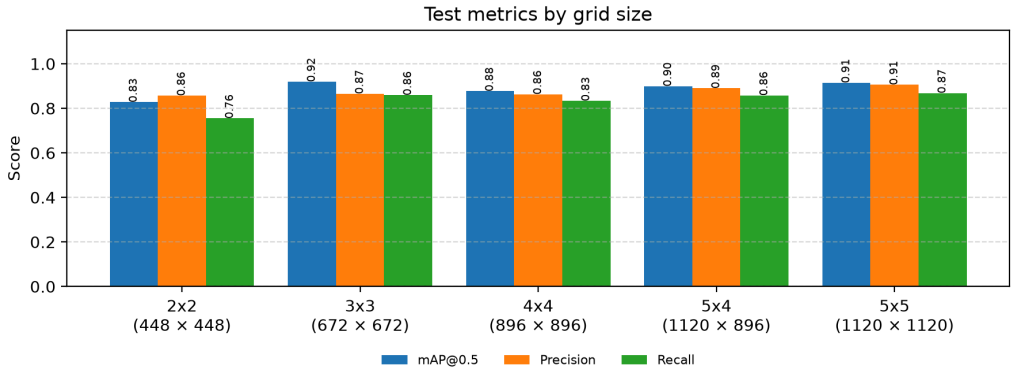


Figure 5. Test mAP@0.5, precision and recall for each grid size (YOLOv8s).

6. Detection Examples

Given a new multi-object image, our script *detect_objects.py* returns the Object ID, name, confidence and bounding box of every detected object. The table below is the output for test image test_00001 (a 3 × 3 grid with 9 objects); 9 of 10 detections match a ground-truth object with the correct ID.

Object ID	Name	Confidence	Top-left (x, y)	Bottom-right (x, y)	Correct
OBJ038	Perfume bottle	0.30	(0, 0)	(62, 147)	no
OBJ063	Measuring Bottle	0.65	(91, 85)	(111, 138)	yes
OBJ038	Perfume bottle	0.94	(310, 41)	(360, 162)	yes
OBJ032	Sound Card	0.98	(526, 102)	(625, 154)	yes
OBJ030	Tennis racket	0.98	(82, 263)	(134, 355)	yes
OBJ019	Plant	0.97	(225, 224)	(447, 383)	yes
OBJ029	Orange	0.95	(535, 323)	(589, 377)	yes
OBJ046	iWatch	0.98	(46, 505)	(166, 526)	yes
OBJ066	hair oil	0.99	(287, 490)	(378, 605)	yes
OBJ073	Bowl	0.98	(519, 505)	(632, 616)	yes



Figure 6. YOLOv8s detections on a 3×3 test grid.

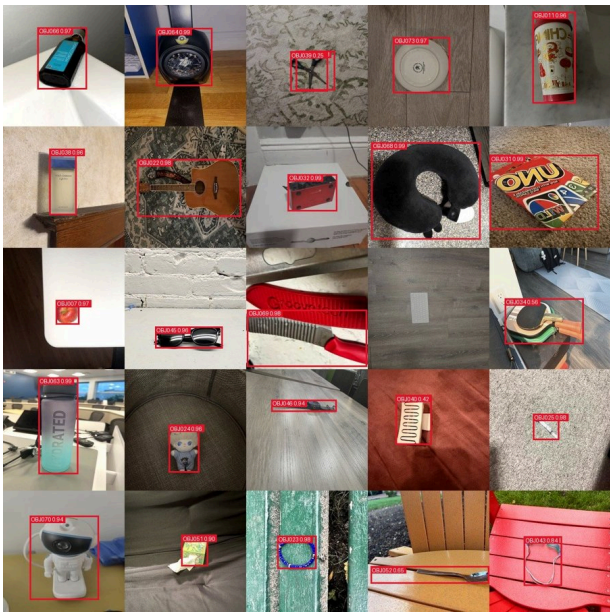


Figure 7. YOLOv8s detections on a 5×5 test grid (25 detections, 25 objects).

7. Comparison and Analysis

Model size.

YOLOv8s has about 3.6 times as many parameters as YOLOv8n. On the test set it reaches mAP@0.5 = 0.893 against 0.877 for YOLOv8n, with precision 0.890 vs. 0.872 and recall 0.856 vs. 0.832. Inference takes 4.6 ms per image for YOLOv8s and 3.9 ms for YOLOv8n on our hardware (Apple M4 GPU, 1120-pixel input).

Grid size.

Test mAP@0.5 ranges from 0.828 on 2x2 grids to 0.920 on 3x3 grids (from best to worst: 3x3 0.920, 5x5 0.915, 5x4 0.899, 4x4 0.878, 2x2 0.828). Somewhat surprisingly, the small 2×2 grids are the hardest. Because every image is resized to 1120 pixels, a 448-pixel 2×2 grid is enlarged about 2.5 times, so its objects appear much larger and blurrier than the objects in the larger grids that make up most of the training data. Overall the differences between grid sizes are small, so the model handles 4 to 25 objects per image.

Per-object results.

32 of the 72 objects reach a test mAP@0.5 of at least 0.99 with YOLOv8s. The weakest objects are Anime Figurine (OBJ044) at 0.12, Computer Keyboard (OBJ012) at 0.35, Sleep Mask (OBJ060) at 0.37, water bottle (OBJ002) at 0.57, Pickleball bat (OBJ049) at 0.58. The most frequent identification errors in the test confusion matrix, counted at the same 0.25 confidence threshold our detection script uses, are: Footstool (OBJ058) predicted as Neck Pillow (OBJ068) (4×); Sunglasses (OBJ014) predicted as Toy Car (OBJ004) (3×); Sleep Mask (OBJ060) predicted as Headphones (OBJ062) (3×); Mobile phone (OBJ013) predicted as Wireless powerbank (OBJ048) (1×).

Error breakdown.

At a confidence threshold of 0.25, of the 2,220 test objects 1,921 (86.5%) were found with the correct ID and a box overlapping the ground truth by IoU ≥ 0.5, 19 (0.9%) were found in the right place but given the wrong ID, and 280 (12.6%) were missed. The model also produced 240 extra boxes that did not match any labelled object, typically duplicates or boxes around a part of an object.

8. Final Model Selection and Justification

We select YOLOv8s as our detection model. It has the highest test mAP@0.5 (0.893) and mAP@0.5:0.95 (0.816) of the models we trained, with precision 0.890 and recall 0.856, and at 4.6 ms per 1120-pixel image it is still fast enough for real-time use.

9. Limitations and Next Steps

- Automatic labels: about 23% of photos were left out because the detectors could not agree on a box. Those photos are typically the hardest ones (small, thin or cluttered objects), so the test set is somewhat easier than the full dataset.
- Some kept boxes come from OWLv2 alone (1,250 photos). They passed a visual spot-check but have not been verified by a second model.
- Grid images are clean: objects never overlap and every object sits in its own cell, unlike a real scene with several objects photographed together.
- A natural next step is testing on real photos of several objects together, and hand-correcting the excluded photos so they can be used.

10. Code Repository

All code for this milestone (labelling, label checking, grid generation, training, evaluation and the detection script) and the trained weights are publicly available in the *milestone2* folder of our GitHub repository. Following the course instructions, no image data is included in the repository.

https://github.com/sankalphegde/ie7615-discriminative-project/tree/main/milestone2